

Algebraic Topology

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1 Introduction

1.1 Recap of IB Topological Spaces

We use algebra to turn topological questions to algebraic questions. This results in a loss of information but lets us prove some results.

Example.

$$\mathbb{R}^n \cong \mathbb{R}^m \implies n = m$$

This is a statement which is very hard to prove otherwise. Likewise we can distinguish topological surfaces with a different number of ‘holes’ in their surface.

For this section all functions are continuous until mentioned otherwise.

Definition. (Map) A *map* is a continuous function.

Lemma. (Gluing lemma) Let $f : X \rightarrow Y$ be a function and $C, K \subset X$ be closed subsets with $C \cup K = X$. Then f is continuous if and only if the restriction of f on C and K is continuous.

Proof. We have that f is continuous if and only if for all closed sets $D \subset Y$ the preimage of D is closed. Since $f^{-1}(D) = (f^{-1}(D) \cap C) \cup (f^{-1}(D) \cap K)$, this holds if and only if $\forall D \subset Y$ closed we have that $f^{-1}(D) \cap C$ and $f^{-1}(D) \cap K$ are closed. Since C and K are closed in X , a subset of C (resp. K) is closed in C (resp. K) if and only if it is closed in X . Hence this holds if and only if $(f|_C)^{-1}(D)$, $(f|_K)^{-1}(D)$ are closed in C and K respectively. Hence this holds if and only if the restriction of f on C and K is continuous. $\square$

Lemma. (Lebesgue number lemma) Let (X, d) be a compact metric space. For any open cover $\mathcal{U} = \{U_\alpha\}_{\alpha \in I}$ of X there exists a $\delta > 0$ such that for all $x \in X$, $B_\delta(x) \subset U_\alpha$ for some $\alpha \in I$.

Proof. Suppose no such δ existed. Then for all $n > 0$ there is some $x_n \in X$ such that $B_{1/n}(x_n) \not\subset U_\alpha$ for all $\alpha \in I$. The sequence $\{x_n\}$ admits a convergent subsequence due to compactness, so wlog replace x_n with its convergent subsequence, and take $x_n \rightarrow y$. Then take $\alpha \in I$ such that $y \in U_\alpha$. Then since U_α is open there exists a $r > 0$ such that $B_r(y) \subset U_\alpha$. Take $N > 0$ such that

$$\begin{cases} \frac{1}{N} < \frac{r}{2} \\ d(x_N, y) < \frac{r}{2} \end{cases}$$

which implies that $B_{1/N}(x_N) \subset B_r(y) \subset U_\alpha$ which is a contradiction. $\square$

1.2 Cell complexes

Definition. (Cell attachment) Given a space X and a map $f : S^{n-1} \rightarrow X$. The space $X \cup_f D^n := X \sqcup D^n / \sim$ where $x \sim f(x), \forall x \in S^{n-1} \subset D^n$ is the result of attaching an n -cell to X along f .

The image of f is called the *attaching sphere*.

Definition. (Cell complex) A *finite-dimensional cell complex* is obtained by:

- (i) Start with a set X^0 with the discrete topology.
- (ii) Inductively once X^{n-1} is defined we form X^n by attaching a collection of n -cells along some collection of attachment maps

$$\{f_\alpha : S^{n-1} \rightarrow X^{n-1}\}_{\alpha \in I}.$$

i.e. that $X^n = (X^{n-1} \sqcup \bigsqcup_{\alpha \in I} D_\alpha^n) / \sim$ where we define $x \sim f_\alpha(x)$ for $x \in S_\alpha^{n-1} = \partial D_\alpha^n$.

- (iii) $X = X^k$ for some $k \geq 0$. The minimal such k is called the dimension of X .

Remark. In general we can make sense of infinite dimensional cell complexes. Suppose that we have

$$X^0 \subset X^1 \subset \dots$$

such that

- (i) X^0 is discrete,
- (ii) X^n is formed by attaching n -cells to X^{n-1} .

Define

$$X = \bigcup_{n \geq 0} X^n$$

with the topology where $U \subset X$ is open $\iff U \cap X^n$ is open in X^n for all n . We say that X is a CW-complex and that X^n is the n -skeleton of X .

2 Homotopy and the fundamental group

2.1 Homotopy

Definition. We denote by I the closed interval $[0, 1]$.

Definition. (Homotopy) Let $f, g : X \rightarrow Y$. Then we say that a map $H : X \times I \rightarrow Y$ is a *homotopy from f to g* if $H(x, 0) = f(x)$ and $H(x, 1) = g(x)$. If such an H exists we say that f is *homotopic* to g and $f \simeq g$.

If $A \subset X$ and $f|_A = g|_A$ then a homotopy H from f to g is relative to A (i.e. a homotopy rel A) if $H(a, t) = f(a) = g(a)$ for all $a \in A$ and $t \in I$. If such an H exists we write $f \simeq_A g$.

Proposition. The relation homotopic rel A defines an equivalence relation.

Proof.

- (i) $f \simeq_A f$ trivially by $H(x, t) = f(x)$.
- (ii) $f \simeq_A g \implies g \simeq_A f$, by replacing t with $1 - t$.

- (iii) Suppose that $f \simeq_A g, g \simeq_A h$. Then let H_1 be the homotopy rel A from f to g and H_2 for g to h . Then define

$$H(x, t) = \begin{cases} H_1(x, 2t) & 0 \leq t \leq \frac{1}{2} \\ H_2(x, 2t - 1) & \frac{1}{2} \leq t \leq 1 \end{cases}.$$

Since $H_1(x, 1) = g(x) = H_2(x, 0)$, H is well-defined, and by the gluing lemma (applied to the closed sets $X \times [0, \frac{1}{2}]$ and $X \times [\frac{1}{2}, 1]$) H is continuous. Also $H(a, t) = f(a)$ for all $a \in A$, so H is a homotopy rel A from f to h and therefore $f \simeq_A h$. $\square$

Definition. (Homotopy equivalence) Given a map $f : X \rightarrow Y$ we say that f is a *homotopy equivalence* if there exists a $g : Y \rightarrow X$ such that $f \circ g \simeq \text{id}_Y$ and $g \circ f \simeq \text{id}_X$. We call g a *homotopy inverse* of f . If there exists a homotopy equivalence $X \rightarrow Y$ then X is *homotopy equivalent* to Y and we write that $X \simeq Y$.

Definition. (Contractible) A space X is *contractible* if X is homotopy equivalent to the one-point space $\{*\}$.

Lemma. Let $f_0, f_1 : X \rightarrow Y$ be homotopic maps and $g_0, g_1 : Y \rightarrow Z$ also be homotopic maps. Then

$$g_0 \circ f_0 \simeq g_1 \circ f_1.$$

Proof. We want to show that $g_0 \circ f_0 \simeq g_0 \circ f_1 \simeq g_1 \circ f_1$.

Choose homotopies H from f_0 to f_1 and H' from g_0 to g_1 . Then $g_0 \circ H$ is a homotopy from $g_0 f_0$ to $g_0 f_1$. We also have that $H' \circ (f_1 \times \text{id}_I)$ is a homotopy from $g_0 f_1$ to $g_1 f_1$ hence we're done by transitivity. $\square$

Proposition. Homotopy equivalence is an equivalence relation.

Proof.

- (i) $X \simeq X$ is obvious from the identity map.
- (ii) $X \simeq Y$ then we have $Y \simeq X$ just by switching the roles of f and g .
- (iii) $X \simeq Y$ and $Y \simeq Z$. Then suppose f, g are the homotopy equivalence and its inverse for X and Y , and f', g' for Y and Z . Consider the map $f'f$ from X to Z and its candidate homotopy inverse gg' . By the lemma above,

$$\begin{aligned} gg'f'f &\simeq g \text{id}_Y f = gf \simeq \text{id}_X \\ f'f gg' &\simeq f' \text{id}_Y g' = f'g' \simeq \text{id}_Z \end{aligned}$$

Therefore $f'f$ is a homotopy equivalence and $X \simeq Z$. $\square$

Example. Consider the inclusion

$$\tau : S^1 \hookrightarrow \mathbb{R}^2 \setminus \{0\}$$

with homotopy inverse $r : \mathbb{R}^2 \setminus \{0\} \rightarrow S^1$ given by projecting each vector onto its unit vector, $r(x) = x/\|x\|$. Since vector normalisation doesn't do anything to unit vectors one of the directions of composition is obvious: $r \circ \tau = \text{id}_{S^1}$. For the other direction,

$$\mathbb{R}^2 \setminus \{0\} \rightarrow S^1 \hookrightarrow \mathbb{R}^2 \setminus \{0\}$$

we're given the homotopy

$$\begin{aligned} H : (\mathbb{R}^2 \setminus \{0\}) \times I &\rightarrow \mathbb{R}^2 \setminus \{0\} \\ (x, t) &\mapsto \frac{x}{t + (1-t)\|x\|} \end{aligned}$$

which is well-defined since the denominator is strictly positive, and is a homotopy from $\tau \circ r$ to $\text{id}_{\mathbb{R}^2 \setminus \{0\}}$. Therefore we have that $S^1 \simeq \mathbb{R}^2 \setminus \{0\}$. Similarly we have that $S^{n-1} \simeq \mathbb{R}^n \setminus \{0\}$.

Example. We have that $\mathbb{R}^n$ is contractible. Let $f : \mathbb{R}^n \rightarrow \{0\}$ be the constant map and $g : \{0\} \hookrightarrow \mathbb{R}^n$ be the inclusion. Then

$$f \circ g = \text{id}_{\{0\}}$$

so one direction is trivial. For the other direction, $g \circ f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is the constant map $x \mapsto 0$. Define the straight-line homotopy

$$\begin{aligned} H : \mathbb{R}^n \times I &\rightarrow \mathbb{R}^n \\ (x, t) &\mapsto tx. \end{aligned}$$

This is continuous since scalar multiplication $\mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ is continuous. We have $H(x, 0) = 0 = (g \circ f)(x)$ and $H(x, 1) = x = \text{id}_{\mathbb{R}^n}(x)$, so H is a homotopy from $g \circ f$ to $\text{id}_{\mathbb{R}^n}$. Therefore $g \circ f \simeq \text{id}_{\mathbb{R}^n}$, so f is a homotopy equivalence with homotopy inverse g . Hence $\mathbb{R}^n \simeq \{0\}$, i.e. $\mathbb{R}^n$ is contractible.

The same argument works for any $X \subset \mathbb{R}^n$ which is star-shaped about some point x_0 (e.g. any convex set), using $H(x, t) = (1-t)x_0 + tx$.